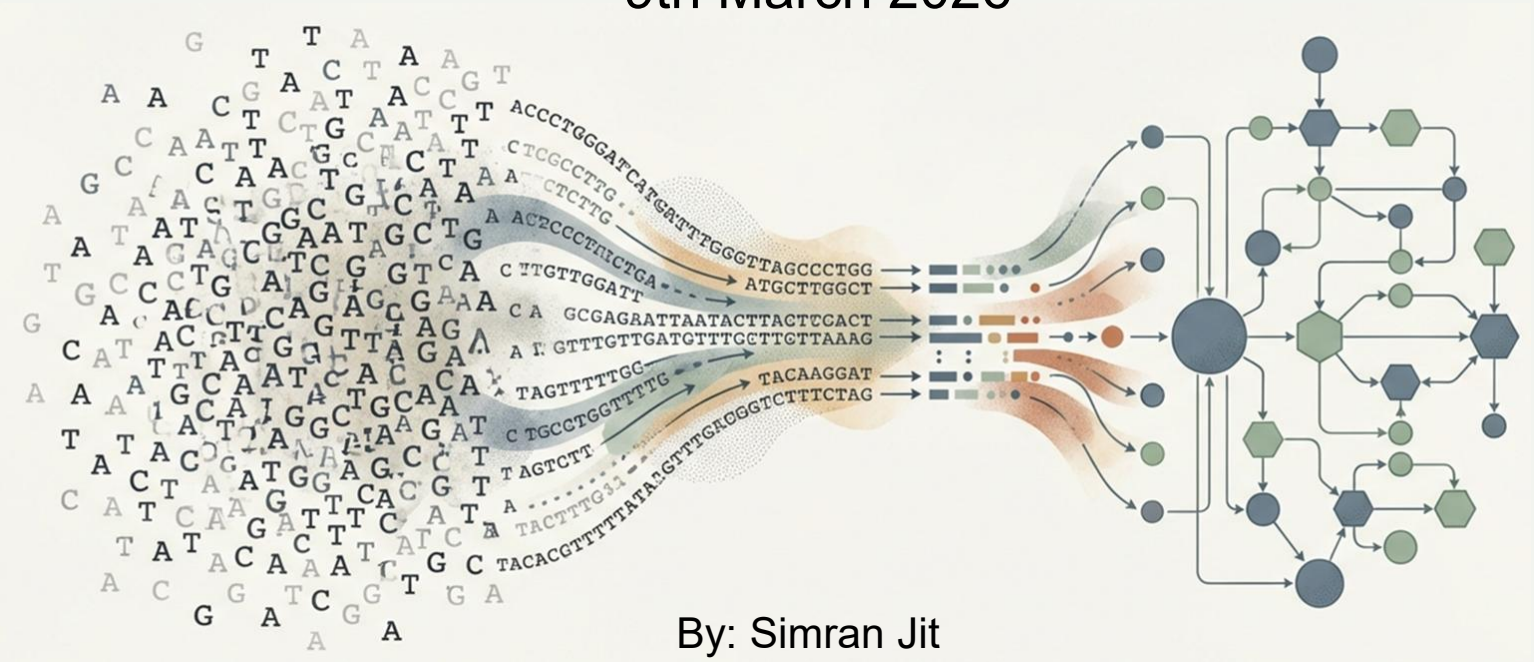
From Raw Reads to Biological Insights

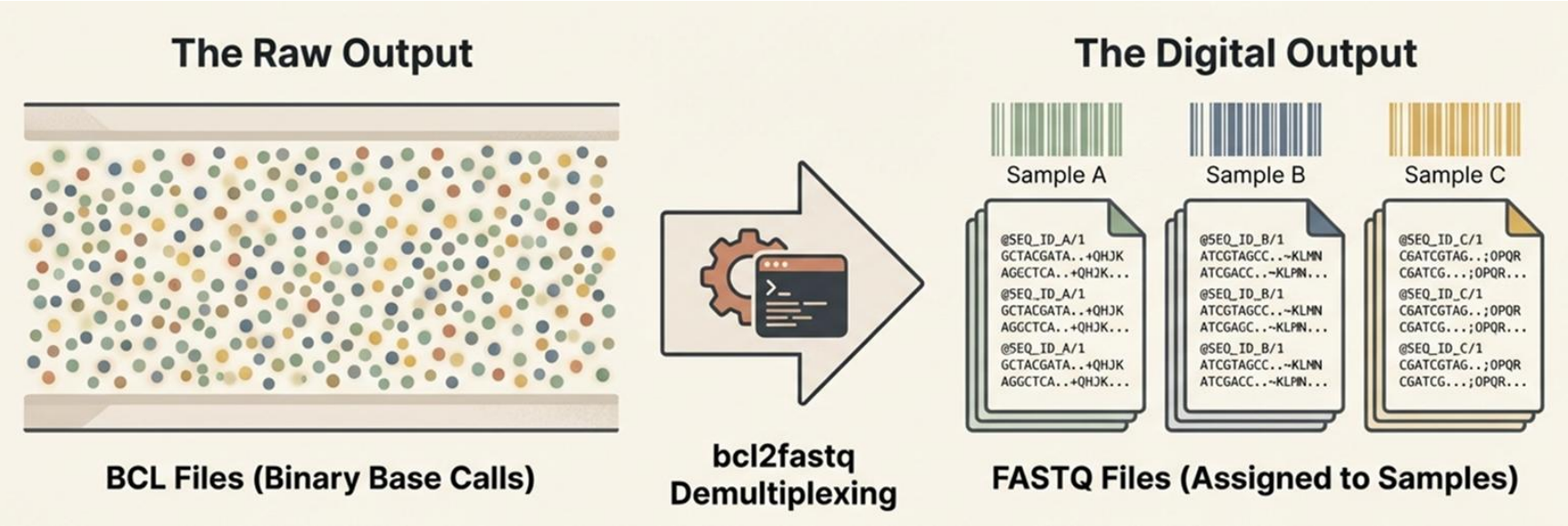
Bulk RNA-seq Data Analysis Workshop

9th March 2026



By: Simran Jit
PhD Scholar
CSIR-IGIB, New Delhi

Sequencers Capture Flashes of Light, Not Biological Letters



- ❑ **Translate light signals to standard digital text.**
- ❑ **Demultiplexing uses index barcodes to assign reads to specific samples.**

Structure of a FASTQ File

Contain sequence & quality scores (Phred scores)

Line 1 (Sequence ID): Starts with @

Line 2 (Nucleotide Sequence): The actual biological data

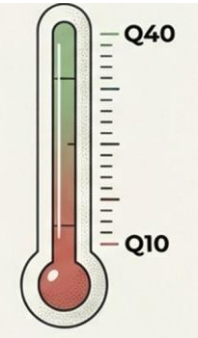
```
@SRR21484222.626.1 626 length=51
GCCTTGGTGGTGAAATGGTAGACTGGAATTCTCGGGTGCCAAGGAACTCCA
+SRR21484222.626.1 626 length=51
F:FFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFF;,FFFFFFFFF
```

Line 3 (Separator): A + symbol, sometimes followed by the ID

Line 4 (Quality Score): ASCII characters representing machine confidence

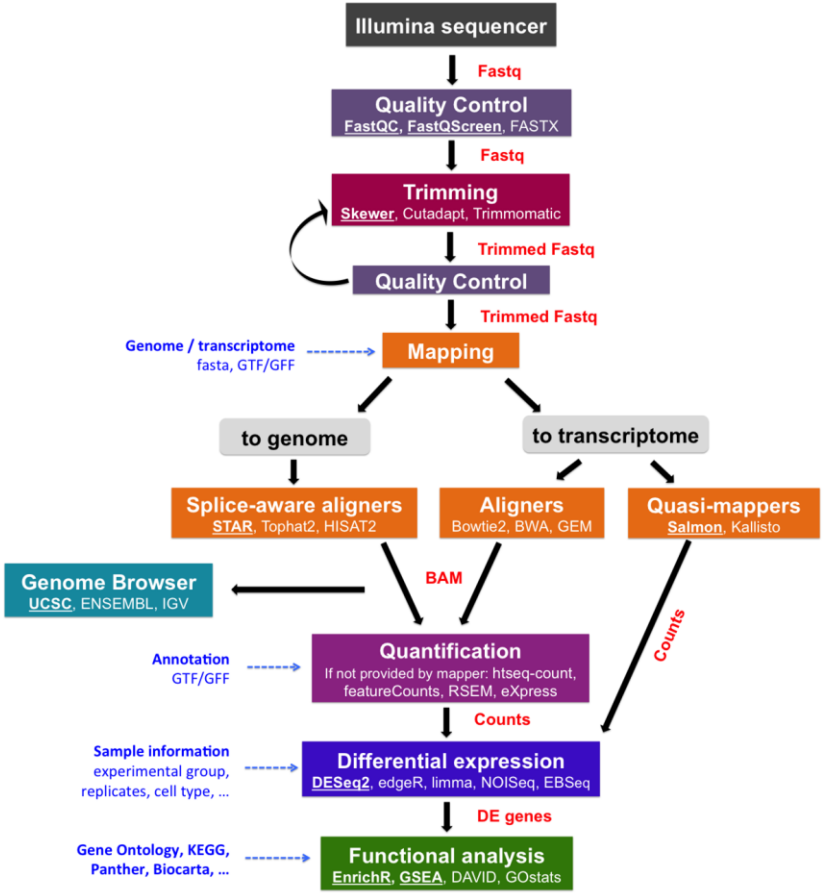
Phred Quality Scores: Interpreting Sequencing Read Quality

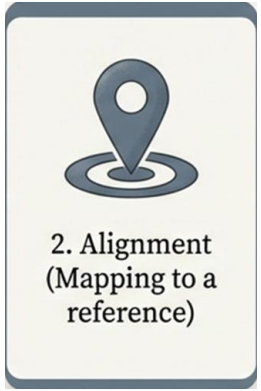
Phred Score	Error Probability / Accuracy
Q10	10% probability of error (90% accuracy)
Q20	1% probability of error (99% accuracy)
Q30	0.1% probability of error (99.9% accuracy)
Q40	0.01% probability of error (99.99% accuracy)

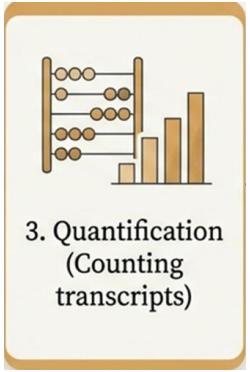


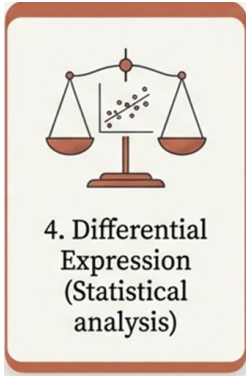
Higher the Phred scores → Lower probability of error

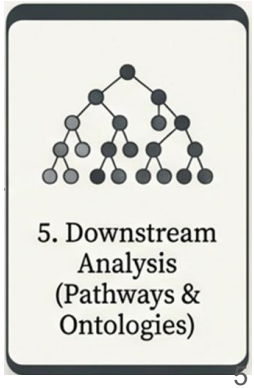
Bulk RNA-seq Data Processing Workflow



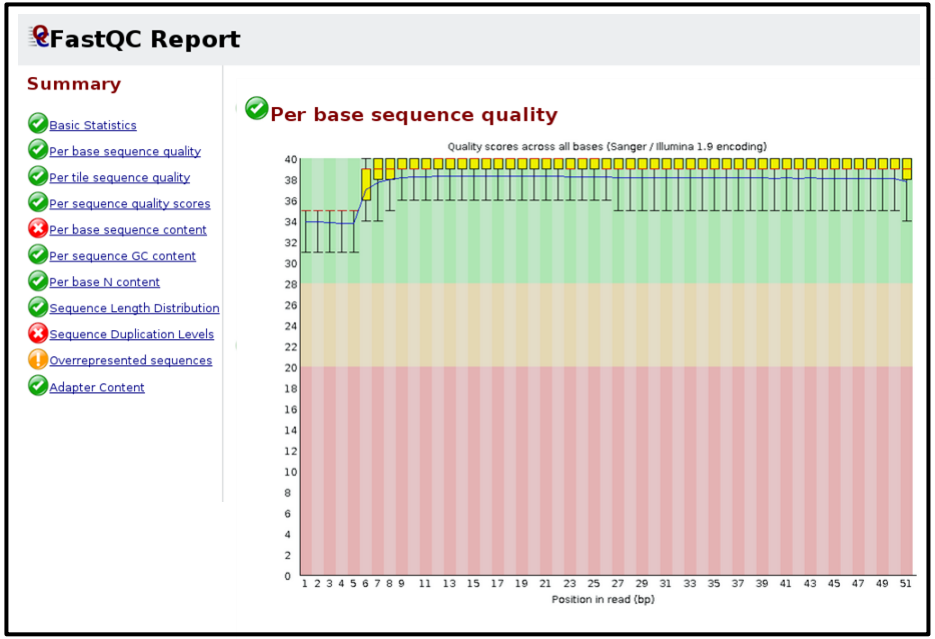


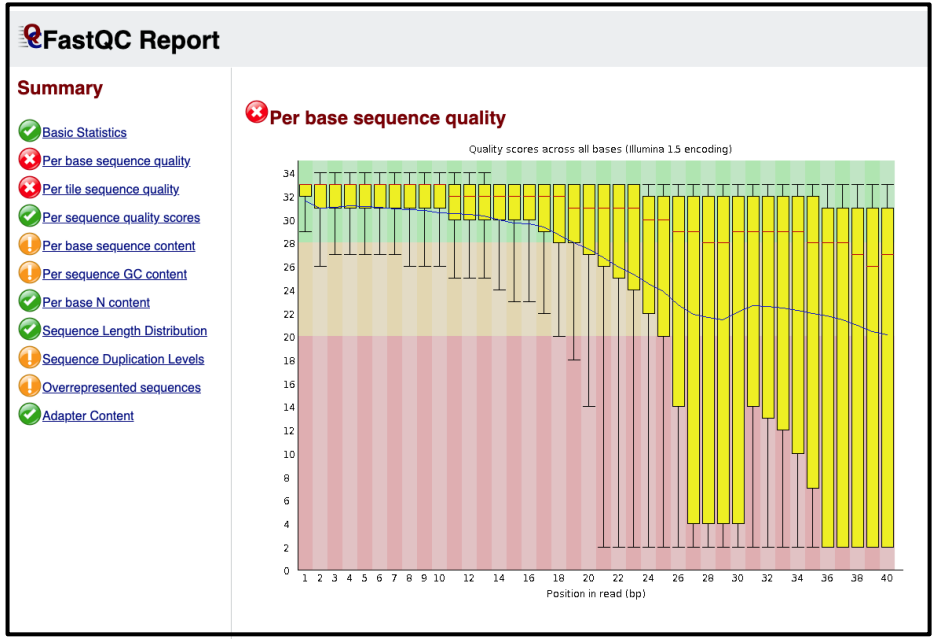




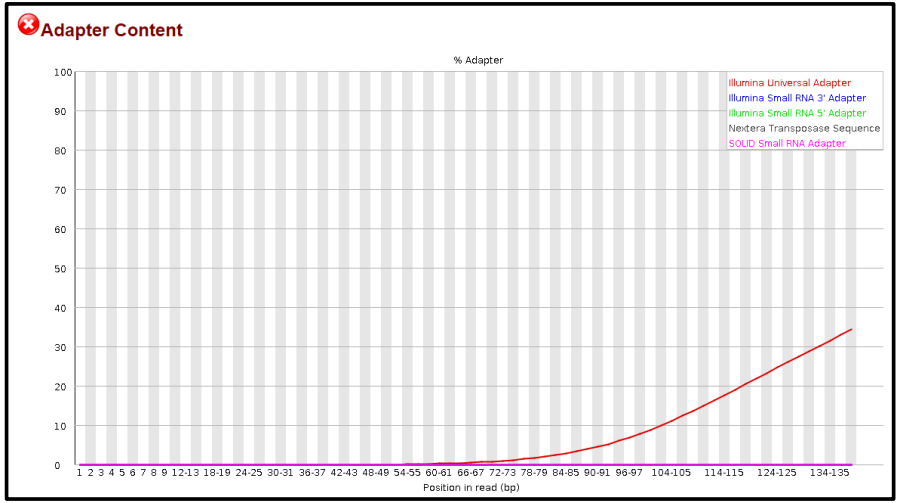


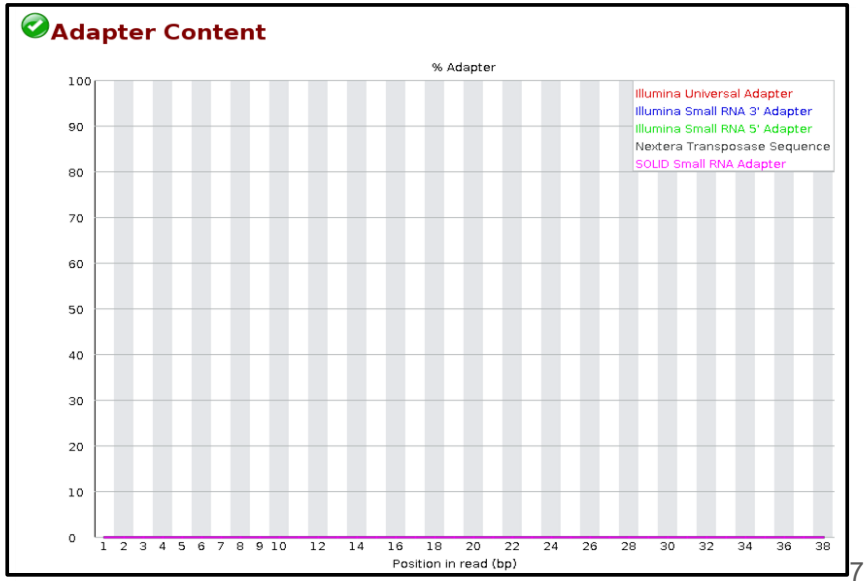
Evaluating Sequencing Read Quality Before Downstream Analysis





Adapter Trimming Improves Read Quality

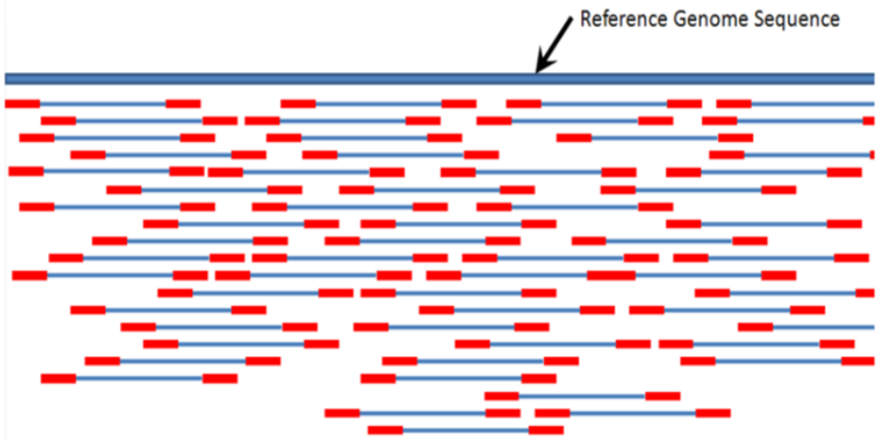




Preprocessing the Reference Genome to Enable Fast Read Alignment

Indexing the genome

Mapping the reads

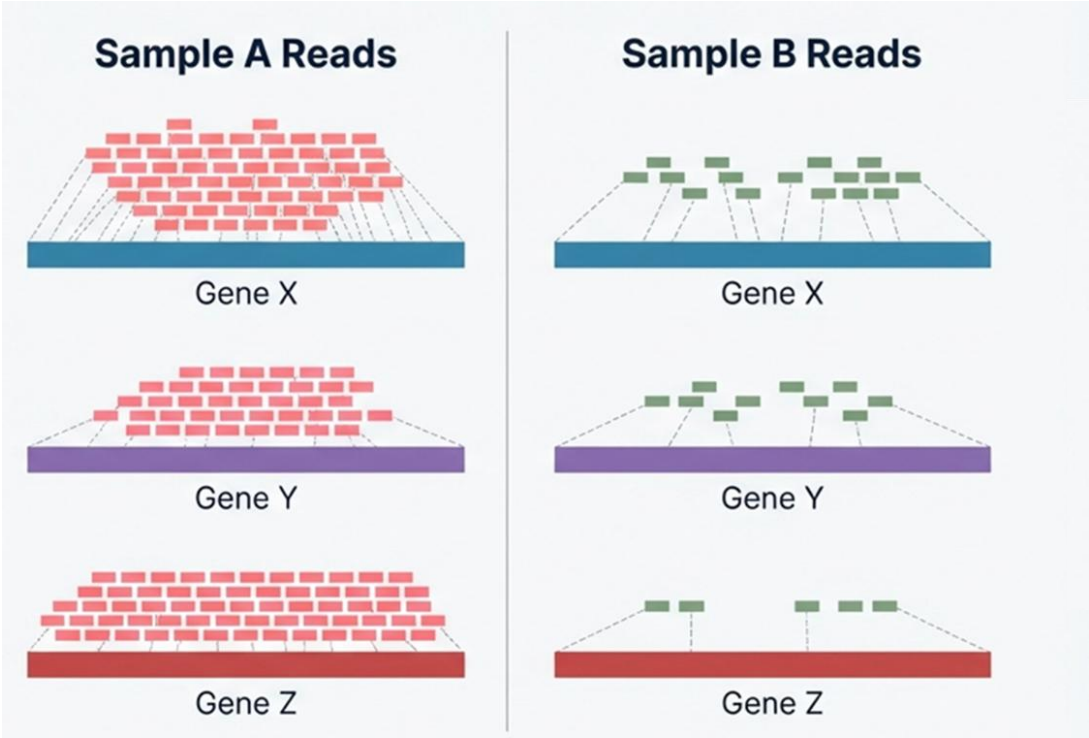


Gene	Sample1	Sample2	Sample3
Gene1	0	10	5
Gene2	14	15	8
Gene3	7	5	2

Raw counts matrix

Normalization Removes Technical Biases so Counts Reflect True Biological Expression

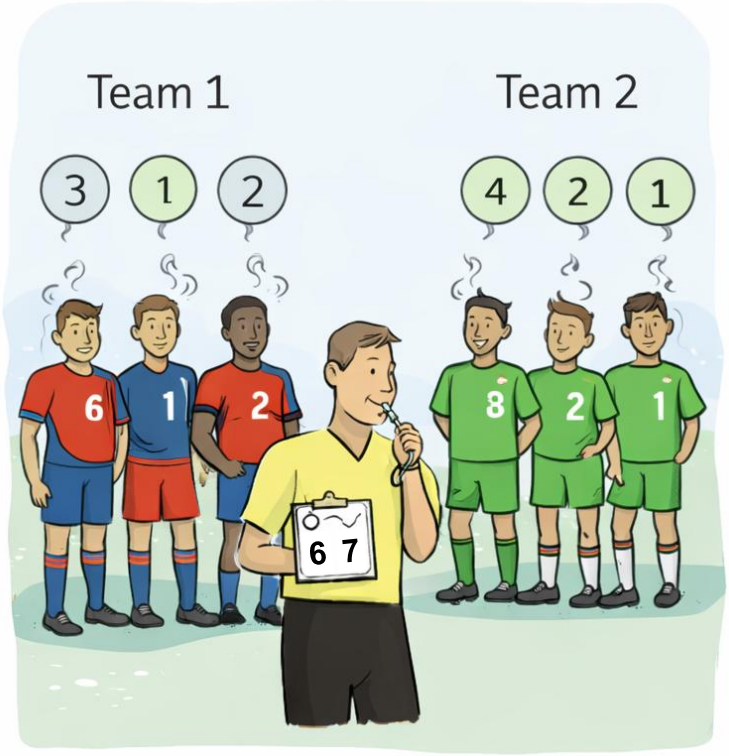
The Challenge: Why Raw Counts Are Misleading



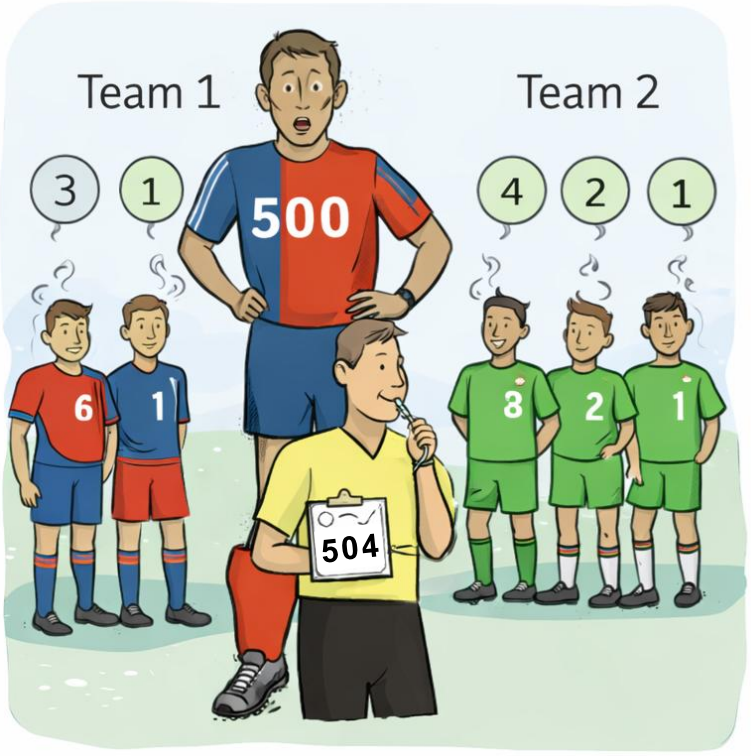
Problem 1: Sequencing Depth
A sample sequenced more deeply will artificially show higher counts for all genes.

Problem 2: Gene Length
Longer genes accumulate more fragments than shorter genes, even if their true expression levels are identical.

Problem 3: RNA Composition
Highly expressed outlier genes can skew the total read distribution of an entire sample.



Fairly equal scores



With Distortion from One Player

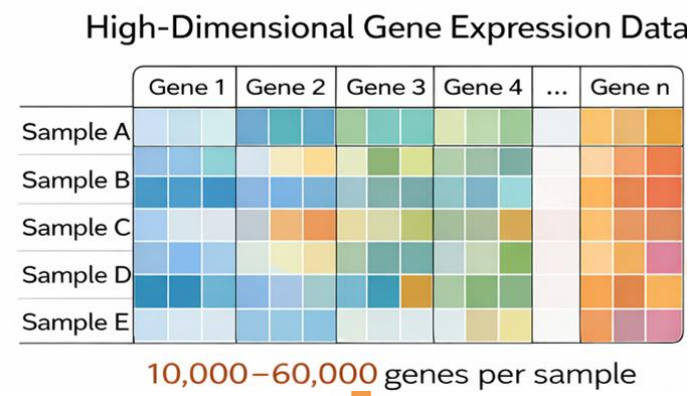
A Glance at RNA-Seq Normalization Methods

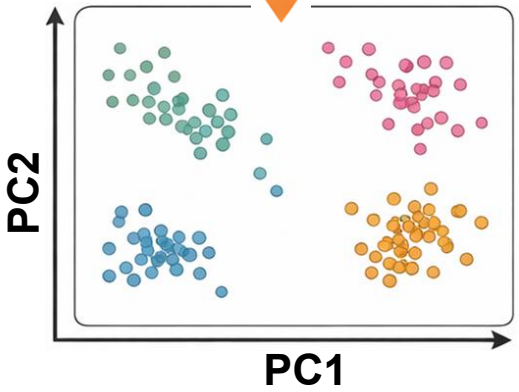
TPM (Transcript per million)	Normalizes by gene length & sequencing depth; Sum of TPMs in each sample equal to 1 million
RPKM/FPKM (Reads/Fragments per kilobase per million)	Normalizes by sequencing depth & gene length
CPM (Counts per million)	Normalizes by sequencing depth
EdgeR's TMM (Trimmed Mean of M-values)	Used in edgeR Accounts for compositional bias (eg. highly expressed genes)
DESeq2 (Median ratio)	Calculates size factors based on median ratio of gene counts to geometric mean across samples Robust against outliers & library size differences

A Guide to Choosing the Right Counts for Your Analysis

Practical Use	Input
Differential Expression Analysis	Feed Raw Counts into DESeq2/edgeR
Expression Level Visualization	TPM or FPKM
Filtering Lowly Expressed Genes	CPM values
Cross-sample Normalization	DESeq2's size factors or edgeR's TMM to adjust for library size & compositional bias

Dimensionality Reduction Summarizes High-dimensional Data for Visualization



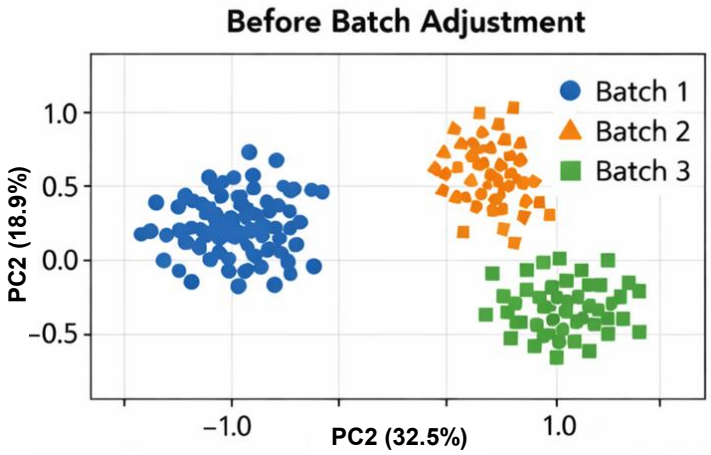


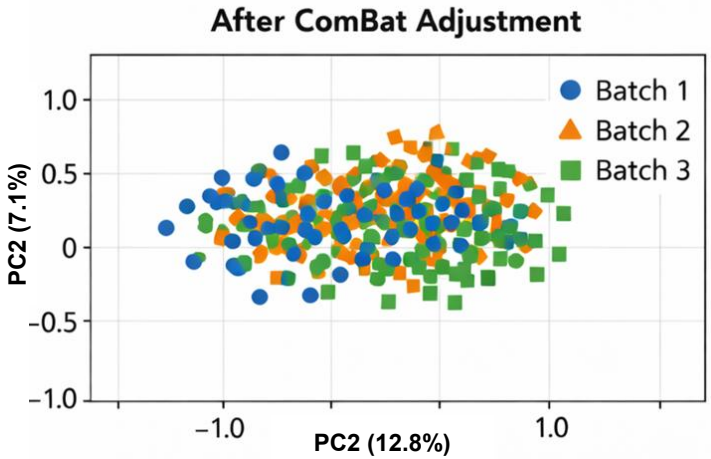
Simplify a complex data matrix into a few "principal components" that capture the most significant variation in the data

Why Principal Component Analysis (PCA)?

Data Visualization and Quality Control: Identifying latent variables → adjustment → reduces statistical noise

Reducing Bias: Identify variables that show batch effects → prevent them from obscuring biological signals & biasing the final conclusions of a study





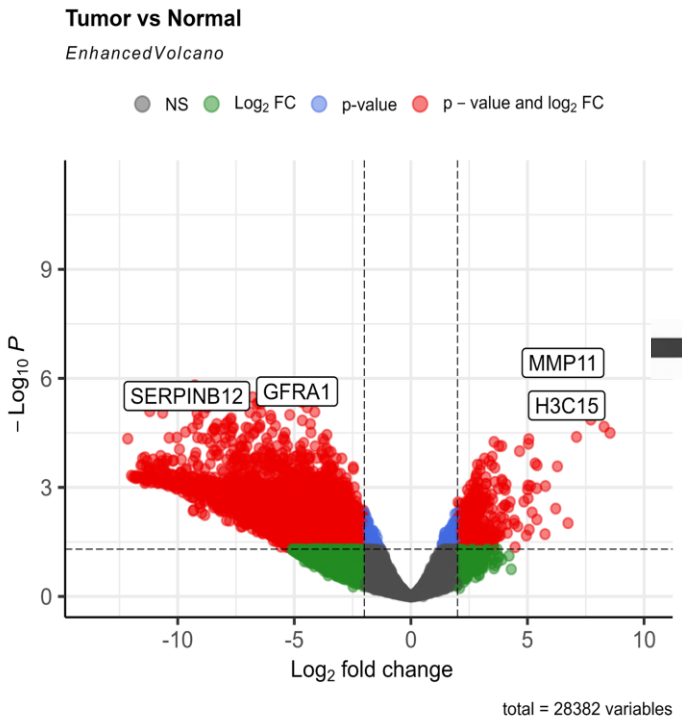
Problem

Samples processed on **different days, in different groups, or by different people** ⟶ major source of **unwanted variation** that can introduce significant bias

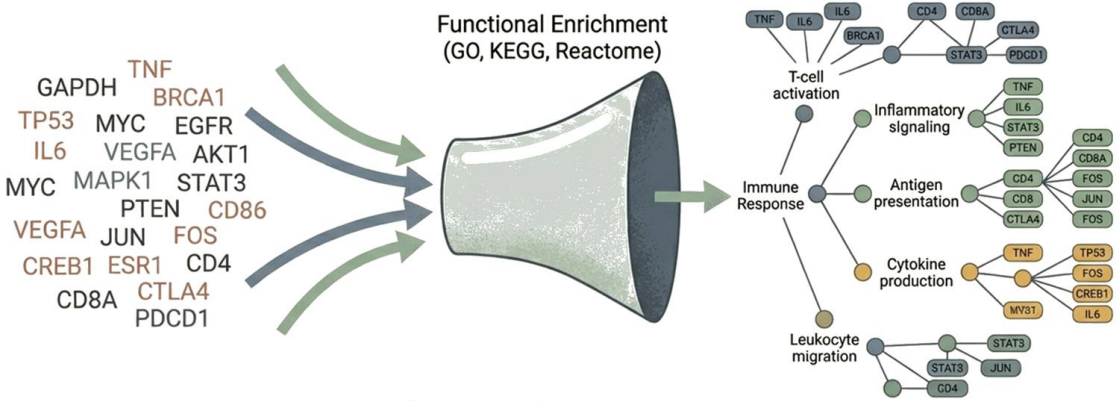
Solution

Batch adjustment using an **empirical Bayesian framework** through ComBat function from sva R package

Differential Expression Reveal Biological Differences; Pathways Reveal Functional Impact



Gene sets from Molecular Signatures Database (*MSigDB*)

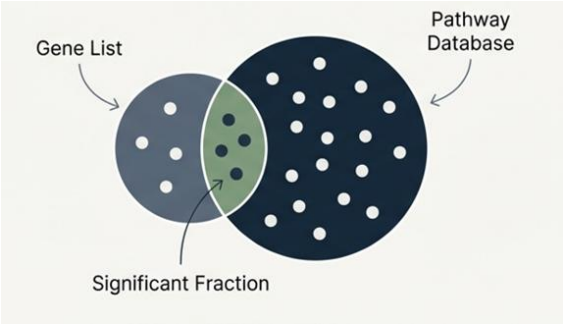


Synthesizing a long list of significant genes into an understandable biological narrative

Pathway Analysis Methods

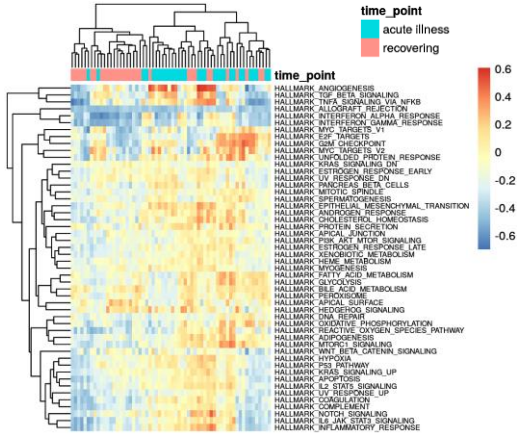
Over-representation Analysis (ORA)

Uses simple 2x2 tables to statistically evaluate the fraction of a gene list found in a known pathway list; hypergeometric test



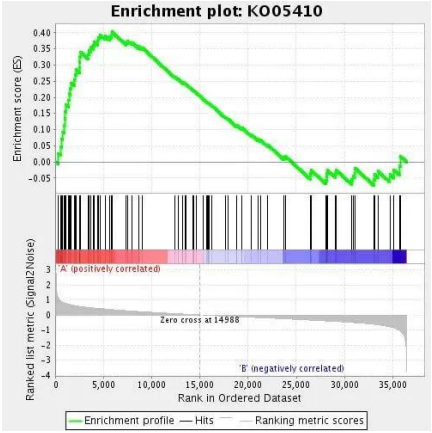
Gene Set Variation Analysis (GSVA)

Gives continuous pathway activity scores per sample; Non-parametric kernel estimation



Gene Set Enrichment Analysis (GSEA)

Evaluates ranked list of genes; generates enrichment scores per gene set; Kolmogorov-Smirnov statistics



Let's Explore Real Data from GEO

Scope: Self Format: HTML Amount: Quick GEO accession: GSE183947 GO

Series GSE183947 Query DataSets for GSE183947

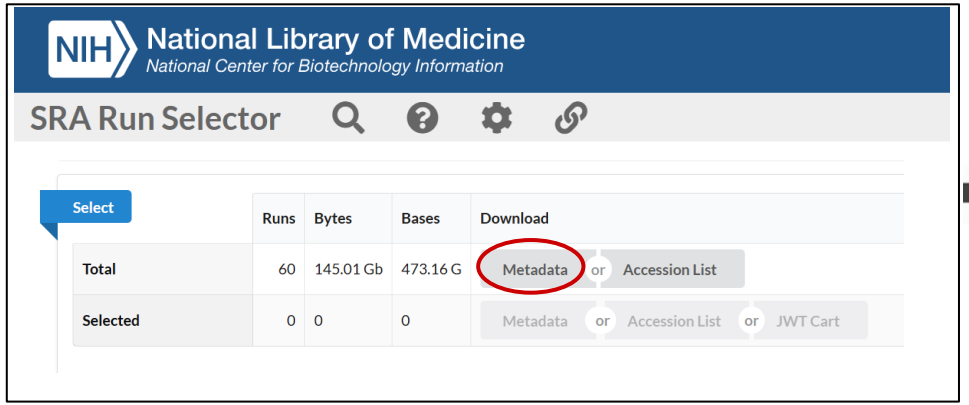
Status	Public on Sep 15, 2021
Title	Identification of five cytotoxicity-related genes involved in the progression of breast cancer
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	Breast cancer is one of the deadly tumors in women, and its incidence continues to increase. This study aimed to identify novel therapeutic molecules using RNA sequencing (RNAseq) data of breast cancer from our hospitals.
Overall design	30 pairs of normal and cancerous tissues from the same excision were collected from the Affiliated Cancer Hospital of Guangzhou Medical University, the Affiliated Cancer Hospital of Sun Yat-sen University and Guangzhou Army General Hospital. RNA sequencing was performed by Guangzhou Huayin Health medical Group. Original reads of RNA sequencing data were normalized as FPKM data.
Contributor(s)	Zhang Y, Wei XX, Yang WK, Shen H
Citation(s)	Zhang Y, Tong GH, Wei XX, Chen HY et al. Identification of Five Cytotoxicity-Related Genes Involved in the Progression of Triple-Negative Breast Cancer. *Front Genet* 2021;12:723477. PMID: 35046993

GSE183947

- ❑ 30 pairs of normal & tumor tissues
- ❑ Processed first 10 pairs of samples

https://github.com/sjigib/bulkrnaseq

Overview of the Steps Followed for Data Processing



SRA Toolkit

FASTQ files

FastQC

Trimmomatic

FASTQC

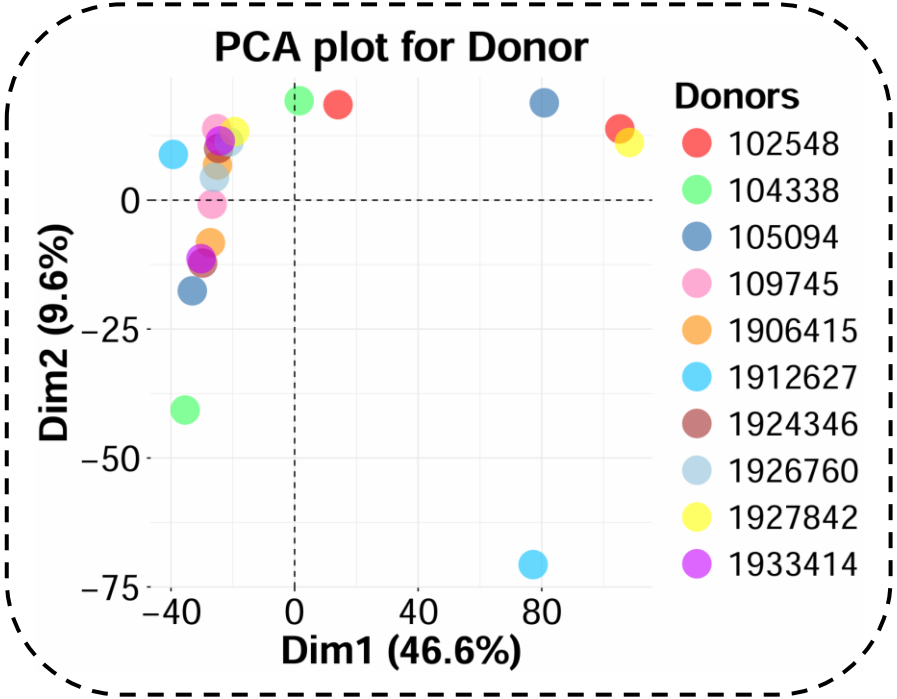
STAR

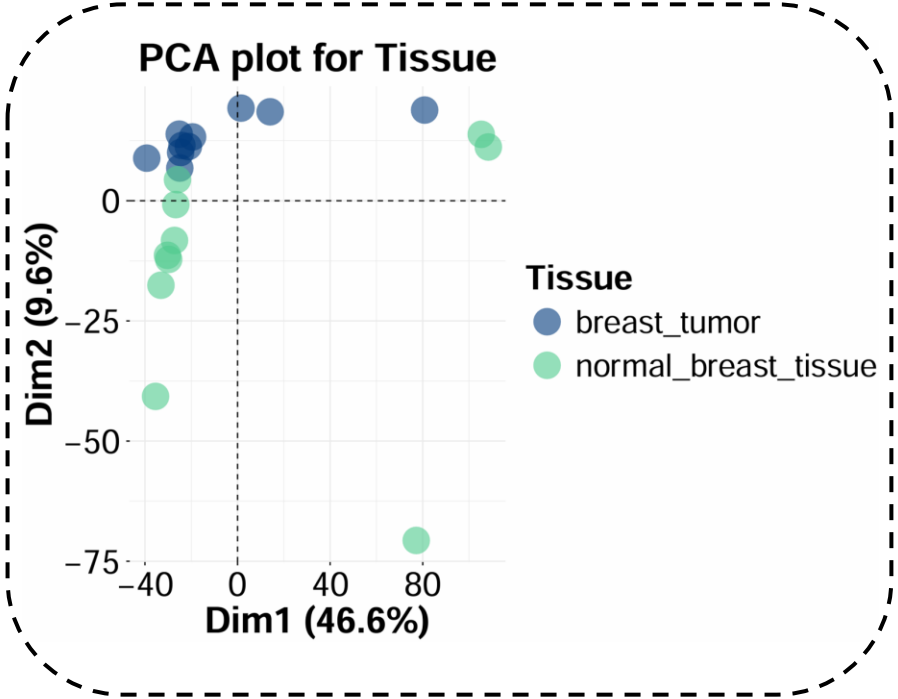
Genome FASTA & GTF file downloaded from Gencode

FeatureCounts

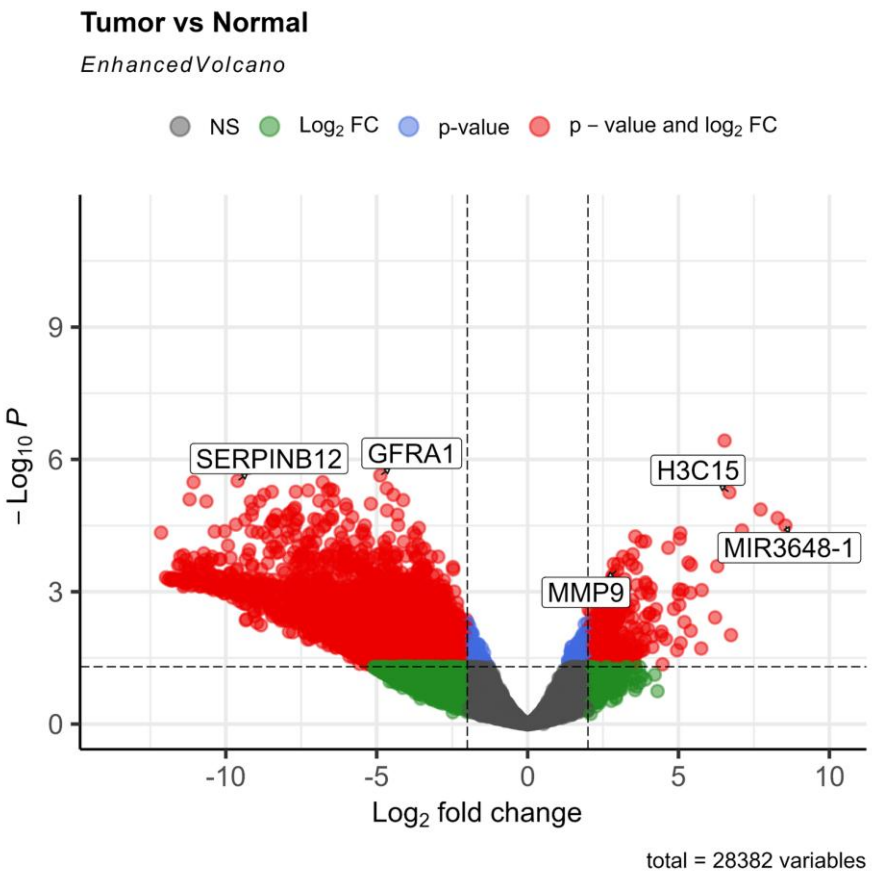
Raw counts matrix

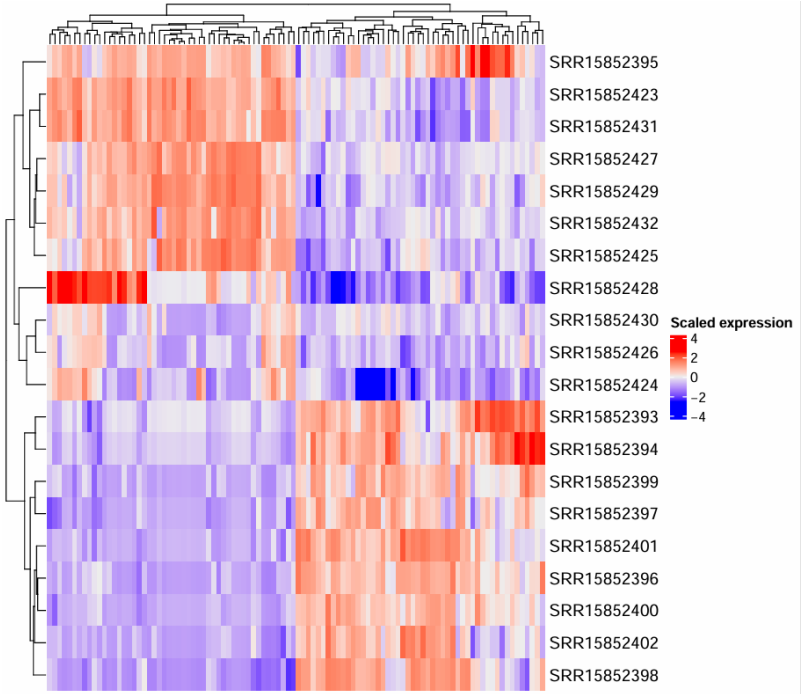
PCA plots (using top 5000 highly variable genes)





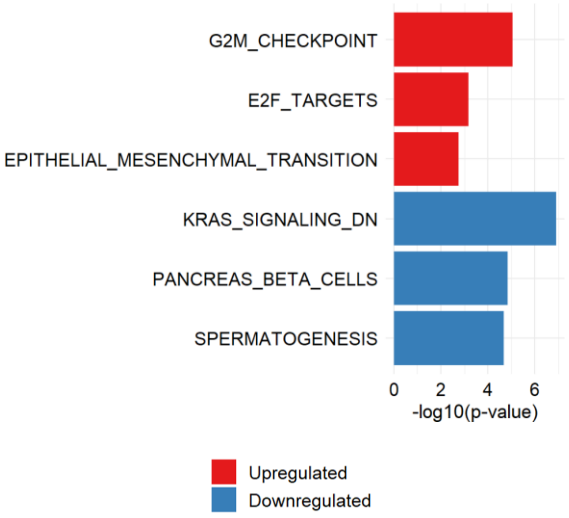
Identification of Differentially Expressed Genes (DEGs)



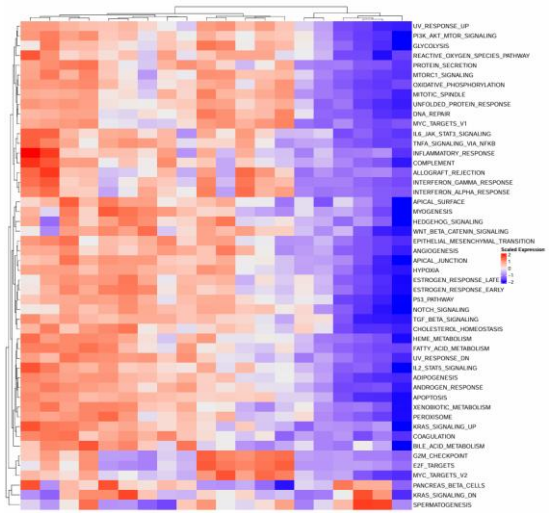


Identifying Key Biological Pathways Altered Between Conditions

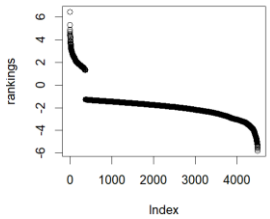
Over-representation Analysis (ORA)

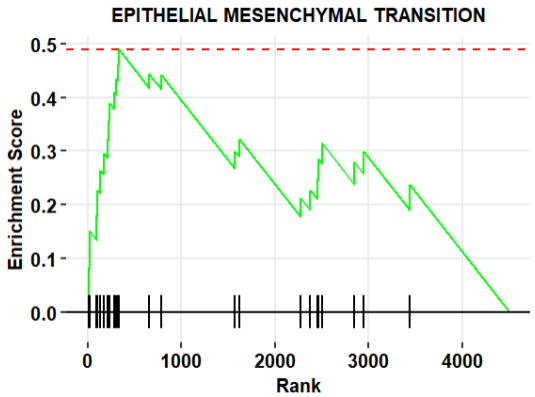


Gene Set Variation Analysis (GSVA)



Gene Set Enrichment Analysis (GSEA)





Thank you!